Project: Data Engineering (DLBDSEDE02)

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Concept: Regional Environmental Monitoring — Stream Processing Pipeline

Data Source

The project uses the **GeoSphere Austria API** as its primary data source. GeoSphere Austria is Austria's national service for geology, geophysics, climatology, and meteorology, responsible for advising the federal government and issuing public warnings across these domains. Its API provides continuous access to real-time measurements from monitoring stations distributed across Austria, including temperature, wind speed, humidity, and air pressure, delivered as structured JSON responses. For this project, stations from multiple small towns within a defined Austrian region are selected as data producers.

Overall Goal

The goal of this data system is to build a scalable, real-time stream processing pipeline that ingests continuous environmental sensor data from multiple regional stations, stores it reliably, and makes it accessible for two key user groups: **regional planners** who require up-to-date and historical environmental data for long-term decision-making, and a **citizen-facing application** that issues warnings when measured values exceed recommended thresholds.

Technology Choice and Justification

Apache Kafka is chosen as the streaming framework. Kafka acts as a distributed message broker between the data producers (API polling scripts per station) and the consumer (database writer). Each station is mapped to a dedicated Kafka topic, enabling parallel ingestion from multiple sources. Kafka ensures reliability through message persistence and fault tolerance, scalability through partition-based horizontal scaling, and maintainability through a clean producer/consumer separation.

MongoDB is chosen as the target database. Since the Geosphere API returns JSON and the regional authority plans to expand its sensor network with yet-unknown data formats, a schema-flexible NoSQL document store is the most appropriate choice. MongoDB supports horizontal scaling and seamless cloud migration, fulfilling the long-term infrastructure requirements.

Alternative considered: Apache Spark Streaming would offer powerful in-stream transformations and aggregations. However, for the primary goal of reliable ingestion and storage of raw sensor data, Kafka's lighter architecture is more appropriate. Spark would introduce unnecessary complexity at this stage.

Implementation Plan

1. Set up a Docker Compose environment with Kafka (+ Zookeeper) and MongoDB containers
2. Implement a Python producer that polls the Geosphere Austria API per station at defined intervals
3. Implement a Python consumer that reads from Kafka topics and writes to MongoDB
4. Configure Kafka topics (one per region/station)
5. Test end-to-end pipeline with live API data
6. Push all code and Dockerfile to a public GitHub repository